MULTIMODAL AI HACKATHON 2026 — TRACK A

Cardiovascular Risk Visualization & Prediction

Project Status, Milestone Audit & Team Work Distribution

Generated on October 20, 2026 • Team Lead: Tooba

1. Executive Project Summary

This project addresses Track A of the Multimodal AI Hackathon 2026 by delivering an integrated 3D interactive human heart viewer coupled with machine learning models that predict **Overall CAD status** alongside localized stenosis in the three critical coronary vessels (**LAD, LCX, RCA**), backed by transparent SHAP clinical feature attributions.

2. Current Progress Audit (What Has Been Done Till Now)

Component	Status	Key Accomplishments & Implementation Details
Data & Leakage Prevention	COMPLETED	Cleaned UCI Z-Alizadeh Sani dataset (303 records, ~55 features). Isolated LAD, LCX, RCA, and Cath from input features to prevent data leakage.
ML Pipelines & Explainer	COMPLETED	Trained multi-target classification pipelines for CAD, LAD, LCX, and RCA; bundled preprocessor and SHAP TreeExplainer into `cardio_models.joblib`.
FastAPI Microservice	COMPLETED	Developed `server.py` with `/predict` and health endpoints, CORS middleware enabled, and feature attribution calculations.
Interactive Frontend UI	COMPLETED	Built React 19 + Vite dashboard with vitals input form, demo presets, real-time metrics cards, and clinical safety disclaimers.
3D Anatomical Foundation	COMPLETED	Created Three.js + React Three Fiber 3D heart canvas with dynamic vessel risk color-coding (<30% Green, 30–60% Amber, >60% Red) and orbit controls.
System Bug Fixes	COMPLETED	Resolved path ampersand issue, installed all frontend packages, added React 19 Suspense boundaries, and integrated Tailwind CSS styling.

3. Team Work Distribution & Ownership

Shaista

30% RUBRIC

AI/ML Engineer • Model Evaluation & Benchmarking

- **Algorithm Benchmarking:** Benchmark CatBoost, XGBoost, LightGBM, SVC, and ElasticNet.
- **Probability Calibration:** Implement `CalibratedClassifierCV` for reliable clinical risk percentages.
- **Imbalance Handling:** Tune class weights for minority stenotic cases in LCX and RCA.
- **Evaluation Table:** Document 5-fold CV metrics (Accuracy, Precision, Recall, F1, ROC-AUC).

Anamta

35% RUBRIC

Core CS • 3D Heart Model & Project Documentation

- **3D Mesh Sourcing:** Source and integrate open-source heart mesh (`.glb`/`.gltf`).
- **Interactive Raycasting:** Implement click/hover detection on LAD, LCX, and RCA arteries.
- **Vessel Callout Tooltips:** Display anatomical branch info and stenosis probability on click.
- **Documentation Lead:** Author the comprehensive 6-page project submission report.

Anushka

35% RUBRIC

AI/ML • Code Integration & Explainability Pipeline

- **Model Integration:** Synchronize Shaista's best model weights into `server.py`.
- **SHAP Clinical Engine:** Refine patient-level waterfall factor attributions.
- **Patient Presets:** Curate realistic demo clinical profiles in `demo_presets.json`.
- **API Latency:** Validate inference speed and error-handling between frontend and backend.

Tooba (Lead)

COORDINATION

Full-Stack Lead • Architecture, Video & Presentation

- **Architecture & Glue:** Coordinate cross-module integration across the team.
- **UI/UX Polish:** Refine dashboard design, responsive layouts, and loading states.
- **Demo Video:** Record and edit the 3–10 minute workflow demonstration video.
- **Pitch & Slides:** Oversee pitch presentation deck and final submission packaging.

4. Sprint Roadmap (What's Left To Be Done)

- **Phase 1 — Model & Asset Prep:** Shaista generates benchmark metric table; Anamta selects the 3D heart `.glb` model.
- **Phase 2 — System Integration:** Anushka hooks up winning model weights and SHAP outputs into `server.py`; Anamta embeds raycasting in React Three Fiber.
- **Phase 3 — Deliverables & Media:** Anamta finalizes the 6-page documentation report; Tooba records the 3–10 min demo video and coordinates slide preparation.